

LAB-08

BL.EN.U4AIE23021

Spectral Analysis of Speech Using DCT and MFCC Features

1. Objective

- To analyze and compare **energy compaction** properties of **Discrete Fourier Transform (DFT)** and **Discrete Cosine Transform (DCT)**.
- To extract and analyze **Mel-Frequency Cepstral Coefficients (MFCCs)** from a speech signal.
- To understand how MFCCs capture **phonetic variations** in speech.

2. Dataset Description

- Type: Speech signal (.wav)
- Sampling Frequency: **22050 Hz**
- Duration: **4s**

3. Methodology

3.1 Preprocessing

- The speech signal is loaded and normalized.
- A **25 ms frame** is extracted from the signal.
- A **Hamming window** is applied to reduce spectral leakage.

3.2 DFT and DCT Analysis

- **DFT** is computed using Fast Fourier Transform (FFT).
- **DCT** is computed using type-II cosine transform.
- Energy is calculated as:

$$E = \sum |X(k)|^2$$

- Cumulative energy is used to analyze **energy compaction**.

3.3 MFCC Extraction

- MFCCs are computed using:
 - Framing
 - Windowing
 - FFT
 - Mel filter banks
 - Log scaling
 - DCT
- Number of coefficients: **13 MFCCs per frame**

4. Results

4.1 Energy Compaction Comparison

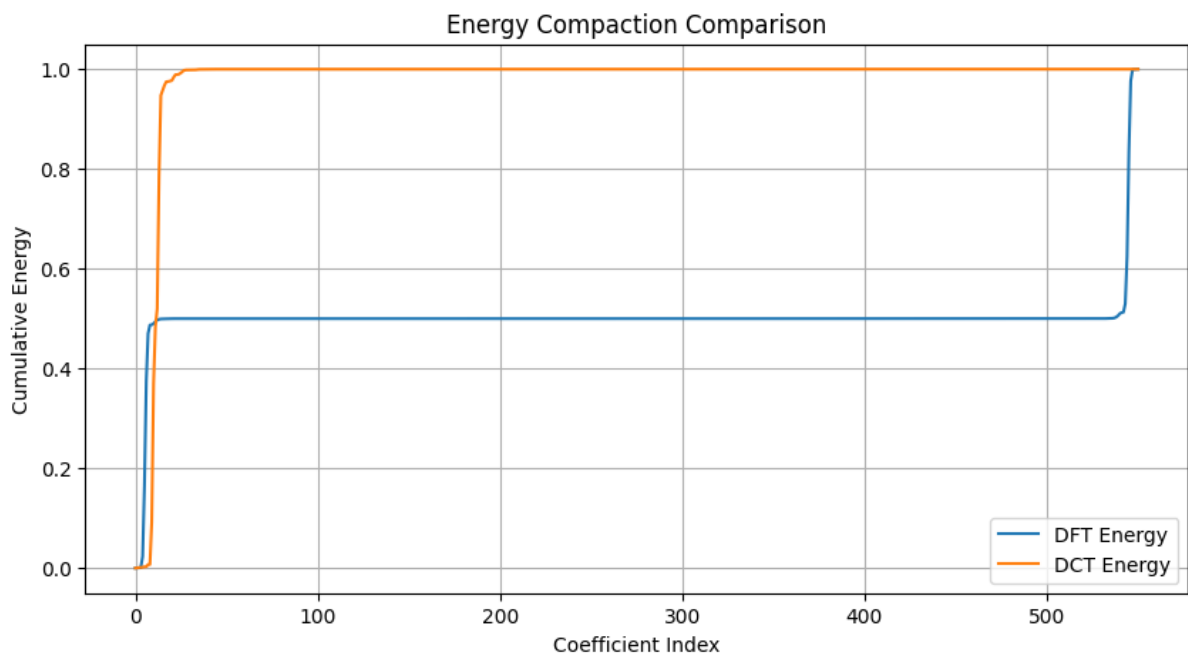


Figure 1: Energy compaction comparison between DFT and DCT.

4.2 Histogram of Coefficients

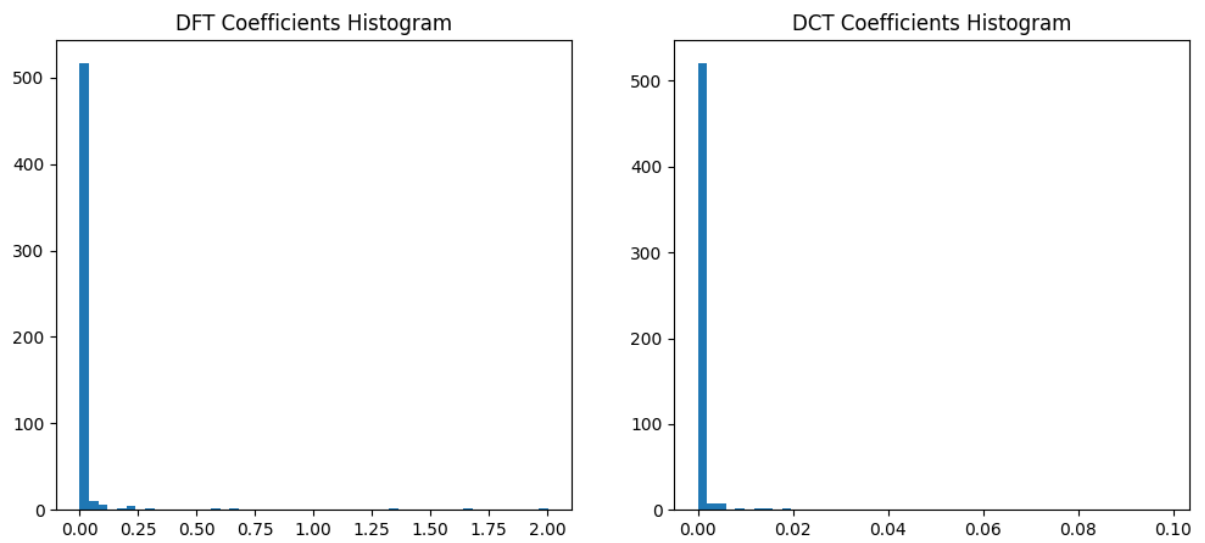


Figure 2: Histogram comparison of DFT and DCT coefficients.

4.3 MFCC Coefficients Over Time

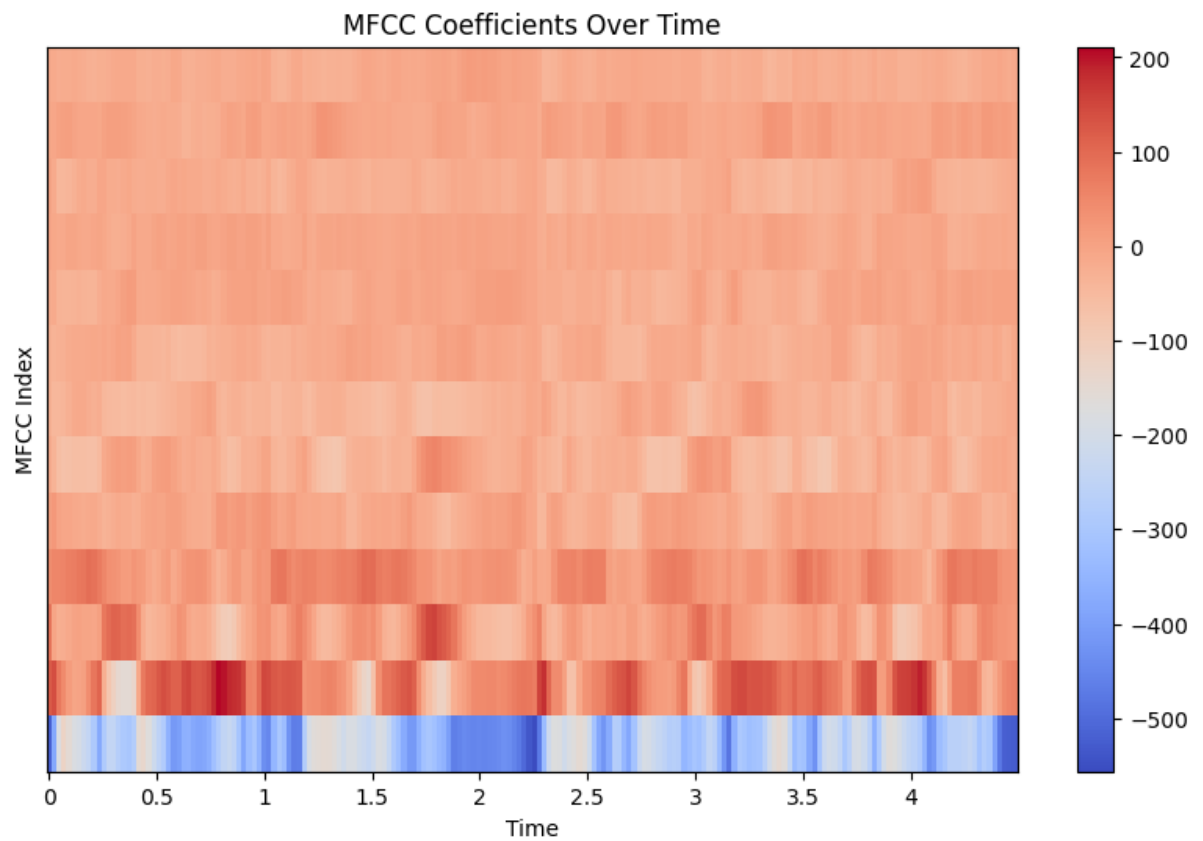


Figure 3: MFCC coefficients variation over time.

4.4 First Two Frames MFCC Comparison

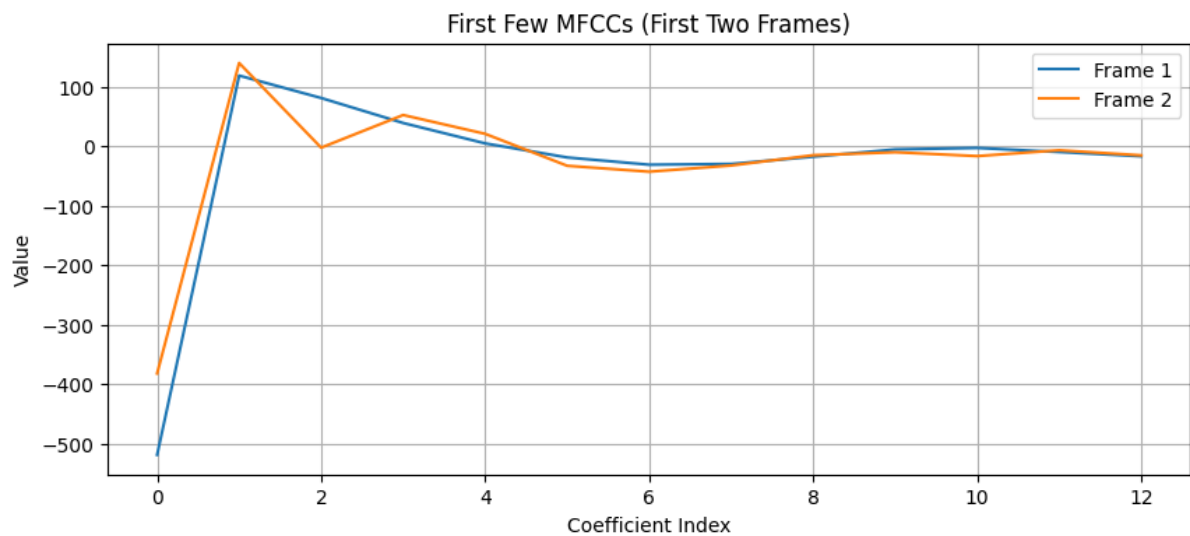


Figure 4: Comparison of MFCC coefficients for first two frames.

5. Discussion

5.1 DCT vs DFT Energy Compaction

The results clearly show that DCT achieves superior energy compaction compared to DFT. Most of the signal energy in DCT is concentrated within the first few coefficients, whereas DFT distributes energy across a larger number of coefficients.

This occurs because:

- DCT uses only cosine basis functions
- DFT includes redundant information due to symmetry

Thus, DCT provides a more efficient representation of speech signals.

5.2 Histogram Analysis

The histogram demonstrates that:

- DCT coefficients are highly concentrated near zero, indicating sparsity
- DFT coefficients are more widely distributed

This confirms that DCT is more suitable for compression and feature extraction tasks.

5.3 MFCC Analysis

The MFCC heatmap shows that coefficients vary over time, reflecting the non-stationary nature of speech.

- **Lower-order MFCCs capture spectral envelope**
- **Higher-order MFCCs capture fine spectral details**

The variation in MFCCs corresponds to changes in phonemes and articulation.

5.4 Frame-wise MFCC Comparison

The comparison of the first two frames shows:

- **Significant variation in lower coefficients**
- **Similarity in higher coefficients**

This indicates that:

- **Lower coefficients carry most of the phonetic information**
- **Speech characteristics change even across short durations**

6. Conclusion

- **DCT provides better energy compaction than DFT, making it more suitable for speech processing.**
- **MFCCs effectively capture the perceptual and phonetic features of speech.**
- **The experiment validates the importance of DCT-based feature extraction in applications like speech recognition.**